Alzheimer's Disease Risk Prediction System

AI-Powered Clinical Decision Support

94.75%	97.31%	99.7%
Accuracy	ROC-AUC	Cost Reduction
15ms	2,149	6 Features
Prediction Speed	Patients	Advanced Tools

DEPI Graduation Project
Data Science & AI Track
November 2025

Team Members:
Shady Kishk (Team Leader), Mostafa Ali, Kirolos Emad, Mohamed Sherif, Amr Ahmed

■ The Problem

Global Healthcare Challenge:

- Over 50 million people worldwide suffer from dementia
- Alzheimer's disease is the most common form (60-70% of cases)
- Early detection is critical but current methods are expensive and inaccessible

Current Diagnostic Barriers:

- MRI/PET brain scans cost $1,000-$3,000 per patient
- Requires specialized equipment and neurologist consultations
- 6-12 month waiting times in many healthcare systems
- Limited availability in rural and underserved areas
- Results in late-stage diagnosis when treatment is less effective

■ Our Solution

We developed a machine learning system that predicts Alzheimer's disease risk using only routinely collected clinical data - **no expensive imaging required**.

■ Key Innovation ■
Uses 32 simple health measurements already collected during routine checkups
Provides instant predictions in under 2 seconds
Costs less than $10/month to operate
Accessible to any clinic with internet connection
Explains predictions with visual AI interpretations

■ How It Works

Step 1: Data Collection

Healthcare providers enter patient information: demographics, lifestyle factors, medical history, and simple cognitive tests.

Step 2: AI Analysis

Our Random Forest machine learning model analyzes all 32 factors simultaneously to identify patterns associated with Alzheimer's disease.

Step 3: Risk Assessment

The system provides risk percentage (0-100%), color-coded risk level (Low/Moderate/High/Critical), visual explanations, and clinical recommendations.

Step 4: Explanation & Reporting

SHAP charts show exactly why the prediction was made, downloadable PDF reports for medical records, and AI chatbot answers questions in real-time.

■ Key Features

■ Real-Time Prediction

Enter patient data → Get instant risk assessment in 15 milliseconds

■ Explainable AI

SHAP waterfall plots show which factors increased or decreased risk

■ Medical Chatbot

Google Gemini AI-powered assistant answers questions about Alzheimer's

■ PDF Auto-Fill

Upload existing medical records → System automatically extracts data

■ Clinical Reports

Generate professional PDF reports with risk assessment and recommendations

■ User-Friendly Interface

Dark/light mode, mobile-responsive design, intuitive sections

■ Project Results

Metric	Target	Achieved	Status
Accuracy	≥90%	94.75%	■ +4.75%
ROC-AUC	≥95%	97.31%	■ +2.31%
Precision	-	92.3%	■ Excellent
Recall	-	89.7%	■ Excellent
Prediction Speed	≤50ms	15ms	■ 70% faster

Cost & Time Savings:

- **99.7% cost reduction:** $1,500 average → $3.50 per assessment
- **99.99% time reduction:** 6 months → 15 seconds
- **2,000x faster** than traditional diagnostic pathways

■ Target Users

Primary Care Physicians: Quick screening tool during routine checkups

Memory Clinics: High-volume risk assessment for patient prioritization

Neurologists: Diagnostic support with transparent AI explanations

Healthcare Administrators: Cost-effective alternative to expensive imaging

Researchers: Validated ML pipeline for Alzheimer's prediction studies

■ Real-World Impact

Democratized Access: No expensive equipment needed, works in any clinic with internet

Earlier Detection: Catches patients before advanced symptoms, allows lifestyle interventions

Empowered Physicians: Primary care doctors can screen confidently with transparent AI

Cost-Effective Healthcare: Reduces unnecessary imaging, prioritizes specialist appointments

■ Team & Contributions

- **Shady Kishk** - Team Leader & Dataset Specialist: Project coordination, dataset specification
- **Mostafa Ali** - Data Scientist: EDA with 18+ visualizations, statistical hypothesis testing
- **Kirolos Emad** - ML Engineer: ML pipeline, model optimization, achieved 94.75% accuracy
- **Mohamed Sherif** - Validation Lead: Rigorous 6-component validation framework
- **Amr Ahmed** - Deployment Engineer: Production web app with 6 advanced features

■ Future Vision

Short-Term (Next 6 Months):

- Clinical trial validation with partner hospitals
- Multi-language support (Arabic, Spanish, French)
- Mobile app versions (iOS/Android)
- Enhanced longitudinal tracking

Long-Term (1-2 Years):

- Multi-class severity staging (Mild/Moderate/Severe)
- Integration with Electronic Health Records (EHR)
- Progression rate prediction
- FDA approval pathway (Class II medical device)

Ultimate Goal: Transform this from a research project into a clinically validated, FDA-approved medical device that democratizes access to early Alzheimer's screening worldwide.

■ Why This Matters

Alzheimer's disease affects millions of families worldwide. Early detection enables lifestyle interventions that slow progression, allows families to plan for future care needs, improves patient outcomes through early treatment, and reduces societal healthcare costs.

Our project demonstrates that **accessible, affordable, and accurate** Alzheimer's screening is possible using only routine clinical data and modern AI techniques.

■ Contact & Resources

- **Project Team:** shadykishk77@gmail.com (Team Leader)
- **GitHub:** https://github.com/ShadyKishk77/Alzheimer-Risk-Prediction
- **Institution:** DEPI (Digital Egypt Pioneers Initiative)
- **Program:** Data Science & AI Track - Graduation Project
- **Date:** November 2025

of cutting-edge AI technology and real-world healthcare needs, demonstrating how data science can address critica